



Constructing Bayesian tangency portfolios under short-selling restrictions

Olha Bodnar^a, Taras Bodnar^b, Vilhelm Niklasson^{b,*}

^a Unit of Statistics, School of Business, Örebro University, 701 82 Örebro, Sweden

^b Department of Mathematics, Stockholm University, 106 91 Stockholm, Sweden

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ABSTRACT

We address the challenge of constructing tangency portfolios in the context of short-selling restrictions. Utilizing Bayesian techniques, we reparameterize the asset return model, enabling direct determination of priors for the tangency portfolio weights. This facilitates the integration of non-negative weight constraints into an investor's prior beliefs, resulting in a posterior distribution focused exclusively on non-negative values. Portfolio weight estimators are subsequently derived via the Markov Chain Monte Carlo (MCMC) methodology. Our novel Bayesian approach is empirically illustrated using the most significant stocks in the S&P 500 index. The method showcases promising results in terms of risk-adjusted returns and interpretability.

1. Introduction

Mean–variance optimal portfolio theory, rooted in the pioneering work of Markowitz (1952), is a cornerstone of modern finance. This theory facilitates the construction of portfolios that maximize expected return for a given risk or, conversely, minimize risk for a predetermined expected return.

Despite its theoretical elegance, direct application of the mean–variance portfolio theory is challenging, as it requires knowledge of the mean vector and covariance matrix of asset returns—values that are intrinsically uncertain. This challenge of parameter uncertainty has been extensively explored in literature, from both frequentist (Jobson and Korkie, 1980; Gibbons et al., 1989; Okhrin and Schmid, 2006; Lassance et al., 2023) and Bayesian statistical perspectives (Zellner and Chetty, 1965; Bawa et al., 1979; Tu and Zhou, 2010; Bodnar et al., 2022), with many recent investigations focused on high-dimensional portfolio selection problems (Bodnar et al., 2018; Cai et al., 2020; Moura et al., 2020; Bodnar et al., 2021; Lassance et al., 2022; Bodnar et al., 2023b; Ding et al., 2023).

The appeal of Bayesian portfolio selection has grown notably during the last decades. Unlike frequentist methods, Bayesian statistics seamlessly integrates prior information, such as investor beliefs or market data, directly into the portfolio optimization process (Black and Litterman, 1992; Rachev et al., 2008). Another advantage of Bayesian methods is their ability to incorporate both parameter and model uncertainties in decision-making, a critical aspect often overlooked by traditional methods. This oversight can lead to suboptimal solutions in conventional approaches by failing to account for all relevant uncertainties in optimization.

The tangency portfolio represents the mean–variance optimal solution when considering the presence of a risk-free asset. This portfolio is identified at the intersection between the mean–variance efficient frontier, depicted as a parabola in the mean–variance diagram (Merton, 1972; Kan and Smith, 2008; Bodnar and Schmid, 2009), and the tangent drawn from the risk-free asset's position. While there is a clear formula for determining the weights of this portfolio when short-selling is permitted, its derivation becomes

* Corresponding author.

E-mail address: niklasson@math.su.se (V. Niklasson).

intricate in the absence of short-selling (Britten-Jones, 1999; Kan and Zhou, 2007). Significant advancements in frequentist mean–variance optimization, particularly with respect to short-selling constraints, have been demonstrated by Jagannathan and Ma (2003) and DeMiguel et al. (2009), who show how imposing specific constraints can enhance portfolio performance.

Despite these frequentist developments, Bayesian estimators of the tangency portfolio offer a different perspective. While several have been explored in the literature (Black and Litterman, 1992; Jorion, 1986; Greysen et al., 2006; Avramov and Zhou, 2010; Tu and Zhou, 2010; Bauder et al., 2018), a common limitation is their lack of constraints on the prior structure, often resulting in portfolios with both positive and negative weights. However, the recent work by Bodnar et al. (2023a) introduces a novel Bayesian approach for the tangency portfolio, enabling the imposition of priors on portfolio weights, and thus offering a new avenue in portfolio optimization.

In this paper, we present an innovative Bayesian methodology for constructing tangency portfolios when short-selling is not allowed, utilizing the parametrization in Bodnar et al. (2023a). Our new approach offers multiple advantages: it facilitates prior specification and inference of the optimal portfolio weights where traditional methods may fall short; it has the potential to reduce portfolio risk by mitigating large negative positions, distinguishing it from existing methods in both Bayesian and frequentist statistics; and it simplifies the computation of portfolio weights using the Markov Chain Monte Carlo (MCMC) approach.

The subsequent sections are organized as follows: Section 2 introduces our innovative Bayesian approach for optimal tangency portfolio construction without short-selling and presents a prior for portfolio weights, resulting in a posterior distribution exclusively concentrated on non-negative values. Section 3 details the empirical study's findings, and Section 4 offers concluding insights.

2. Bayesian tangency portfolio

Let $\tilde{\mathbf{x}}_1, \dots, \tilde{\mathbf{x}}_n$ represent the p -dimensional vectors of asset returns for p assets, and let $r_{f,i}$ denote the return of the risk-free asset at time i . The vectors of excess returns are given by $\mathbf{x}_1 = \tilde{\mathbf{x}}_1 - r_{f,1}\mathbf{1}, \dots, \mathbf{x}_n = \tilde{\mathbf{x}}_n - r_{f,n}\mathbf{1}$ where $\mathbf{1}$ is a p -dimensional vector of ones. By $\mathbf{x}_{1:n} = (\mathbf{x}_1, \dots, \mathbf{x}_n)$ we denote the data matrix which consists of n realizations of the p -dimensional vectors of the excess stock returns.

The vector $\boldsymbol{\mu}$ denotes the expected returns of the excess returns, and $\boldsymbol{\Sigma}$ is their associated covariance matrix. In our Bayesian context, we assume that the vectors of excess returns, given $\boldsymbol{\mu}$ and $\boldsymbol{\Sigma}$, are independent and follow a multivariate normal distribution. However, without conditioning, it is important to emphasize that the asset returns are neither independent nor normally distributed, as noted in Bernardo and Smith (2000) and Bodnar et al. (2022).

Under the given assumptions on the asset returns, it follows that, conditionally on $\boldsymbol{\mu}$ and $\boldsymbol{\Sigma}$, the joint distribution of the asset returns is a member of the exponential family (Sundberg, 2019). The likelihood function can be expressed as

$$\begin{aligned} f(\mathbf{x}_{1:n} | \boldsymbol{\mu}, \boldsymbol{\Sigma}) &\propto \det(\boldsymbol{\Sigma})^{-n/2} \exp\left(-\frac{1}{2} \sum_{i=1}^n (\mathbf{x}_i - \boldsymbol{\mu})^\top \boldsymbol{\Sigma}^{-1} (\mathbf{x}_i - \boldsymbol{\mu})\right) \\ &= \det(\boldsymbol{\Omega})^{n/2} \exp\left(-\frac{n}{2} \mathbf{v}^\top \boldsymbol{\Omega}^{-1} \mathbf{v}\right) \exp\left(\left(\sum_{i=1}^n \mathbf{x}_i\right)^\top \mathbf{v} + \text{tr}\left(\boldsymbol{\Omega} \left(-\frac{1}{2} \sum_{i=1}^n \mathbf{x}_i \mathbf{x}_i^\top\right)\right)\right), \end{aligned} \quad (2.1)$$

where

$$\mathbf{v} = \boldsymbol{\Sigma}^{-1} \boldsymbol{\mu} \quad \text{and} \quad \boldsymbol{\Omega} = \boldsymbol{\Sigma}^{-1} \quad (2.2)$$

are the canonical parameters, and the canonical statistic $(\mathbf{t}, -T/2)$ is given by

$$\mathbf{t} = \sum_{i=1}^n \mathbf{x}_i \quad \text{and} \quad T = \sum_{i=1}^n \mathbf{x}_i \mathbf{x}_i^\top. \quad (2.3)$$

Given the canonical representation of the likelihood function, the optimal portfolio weights based on the mean–variance optimization problem become particularly significant. The problem is formulated as

$$\max_{\mathbf{w}} \left\{ \mathbf{w}^\top \boldsymbol{\mu} - \frac{\gamma}{2} \mathbf{w}^\top \boldsymbol{\Sigma} \mathbf{w} \right\}, \quad (2.4)$$

where $\mathbf{w}$ represents the portfolio weights and γ is the risk aversion coefficient. When there are no short-selling constraints, the optimal weights can be expressed as (see, e.g., Ingersoll, 1987; Britten-Jones, 1999)

$$\mathbf{w} = \gamma^{-1} \boldsymbol{\Sigma}^{-1} \boldsymbol{\mu} = \gamma^{-1} \mathbf{v}, \quad (2.5)$$

i.e., the solution is proportional to the unscaled tangency portfolio weights. Hence the canonical representation allows us to directly model the weights of the tangency portfolio. This not only facilitates the effortless incorporation of investor beliefs regarding these weights into the prior but also pioneers a shift in Bayesian inference procedures, sculpting them around these weights.

As demonstrated in Bodnar et al. (2023a), this novel approach has been successfully applied to model general tangency portfolios, accommodating both long and short positions. The authors elucidated how diverse information sources can be integrated to bolster the efficacy of a conjugate prior. However, it is important to recognize scenarios where investors either prefer portfolios consisting solely of long positions or where short-selling is entirely prohibited in the equity market. Such preferences can be seamlessly addressed by employing specialized priors that exclusively accommodate long positions.

2.1. Bayesian inference

Within our Bayesian framework, to ensure that the tangency portfolio weights remain positive, we use a prior distribution that exclusively supports positive values. This is based on the principle that the support of the posterior cannot exceed the support of the prior. One approach to realize this is by modeling the logarithm of $\mathbf{v}$ as a multivariate normal distribution. Such a modeling ensures that $\mathbf{v}$ only assumes positive values, a technique frequently adopted in optimization.

For the matrix $\mathbf{\Omega}$, which represents the inverse of a covariance matrix, a prior is also necessary. Given its symmetric nature and prevailing practices in modern Bayesian applications, the Lewandowski–Kurowicka–Joe (LKJ) distribution, combined with a random standard deviation, is often utilized.

To be precise, let $\mathcal{MN}$ symbolize the multivariate normal distribution, C^+ denote the Half-Cauchy distribution, and $\mathcal{LKJ}$ represent the LKJ distribution. Based on these notations, our priors are formulated as

$$\log(\mathbf{v}) \sim \mathcal{MN}(\log(\mathbf{w}_0), \kappa \mathbf{I}), \quad (2.6)$$

$$\sigma_i \sim C^+(\beta) \quad \text{for } i = 1, 2, \dots, p, \quad (2.7)$$

$$\mathbf{R} \sim \mathcal{LKJ}(\eta), \quad (2.8)$$

$$\mathbf{\Omega} = \text{diag}(\boldsymbol{\sigma}) \mathbf{R} \text{diag}(\boldsymbol{\sigma}), \quad (2.9)$$

where $\mathbf{I}$ is the identity matrix, κ , β , and η are constants, controlling the shapes and scales of the prior distributions, and $\mathbf{w}_0$ represents the prior belief about the optimal portfolio weights. Notably, the parameter κ has a pivotal role in determining the level of trust in the prior belief about $\mathbf{w}_0$. A larger value of κ indicates a diminished reliance on the prior.

Let $\pi(\mathbf{v}, \mathbf{\Omega})$ denote the prior discussed above. Then the posterior for $\mathbf{v}$ and $\mathbf{\Omega}$ is given by

$$\pi(\mathbf{v}, \mathbf{\Omega} | \mathbf{x}_{1:n}) \propto \det(\mathbf{\Omega})^{n/2} \exp\left(-\frac{n}{2} \mathbf{v}^\top \mathbf{\Omega}^{-1} \mathbf{v}\right) \exp\left(\mathbf{t}^\top \mathbf{v} + \text{tr}\left(-\frac{1}{2} \mathbf{\Omega} \mathbf{T}\right)\right) \pi(\mathbf{v}, \mathbf{\Omega}). \quad (2.10)$$

The influence of the prior diminishes with larger sample sizes, consistent with the Bernstein–von Mises theorem (Bernardo and Smith, 2000). However, our chosen prior is key under short-selling restrictions, ensuring a posterior distribution of portfolio weights that are exclusively positive. Thus, Bayesian estimators based on this posterior will yield only positive tangency portfolio weights. Exploring alternative positive-value priors remains a promising area for future research.

The expression of the posterior (2.10) cannot be directly applied for Bayesian inference procedures for $\mathbf{v}$ because of its complex structure. We address this issue by using the MCMC method to extract samples from the posterior distribution. From these samples, we derive Bayesian point estimators and credible sets for the optimal portfolio weights. We demonstrate this approach in Section 3 with a real-world application.

3. Empirical illustration

This section showcases the application of our novel method (referred to as the Log-Normal Prior Tangency Portfolio) using empirical stock returns. We analyze the top stocks in the S&P 500 index between January 2006 and June 2023, a timeframe capturing significant market fluctuations such as the 2008 Lehman Brothers collapse and the 2020 COVID-19 crisis.

For our analysis, we implement monthly portfolio rebalancing with a rolling window of $n = 60$ months (or five years). Employing low-frequency returns, such as monthly data, tends to better satisfy the assumption of conditional normality (Fama, 1976). The risk-free rate corresponds to the US 3-Month Discounted Treasury Bill (DTB3), adjusted to its bond equivalent yield. Data sources include stock prices from Alpha Vantage, market capitalizations from Financial Modeling Prep, and the S&P 500's historical components, sourced from the GitHub repository <https://github.com/fja05680/sp500>. The data in this repository are associated with, and partly derived from, the material in the book by Clenow (2019). During each rebalancing phase, portfolios are adjusted to include either the top 10 or top 20 largest stocks in the S&P 500 by market capitalization, allowing us to analyze and compare the performance across these two distinct portfolio sizes. This selection, despite the relative ease of short-selling large-cap stocks, aligns with the investment strategies of many funds that are restricted from short-selling, yet seek the stability of highly liquid and low volatility stocks. Moreover, our focus on large-cap stocks is influenced by the availability of reliable data, essential for robust analysis.

In our novel method, we initialize the weights, $\mathbf{w}_0$, to be the value-weighted portfolio, a choice particularly apt for large-cap stocks; for portfolios with a broader range of stock sizes, an alternative initialization of $\mathbf{w}_0$ might be warranted to account for the higher idiosyncratic risk of smaller stocks. Furthermore, we use parameters $\beta = 5$ and $\eta = 2$. To study the method's sensitivity and its adaptation, we experiment with two levels of κ : 1 and 2. A lower value of κ signifies a slightly greater belief in the prior weights.

To comprehensively evaluate our methodology, we compare it against several benchmark strategies, including well-established Bayesian and traditional approaches. Each strategy is subject to monthly rebalancing. The strategies are as follows:

1. **Value-Weighted Portfolio:** Mirrors the S&P 500's broader performance by allocating weights based on market capitalization.
2. **Equally-Weighted Portfolio:** Allocates uniform weights to all assets.
3. **Black–Litterman Portfolio:** Uses market equilibrium for optimal weights calculation, as described in Black and Litterman (1992), with the value-weighted portfolio as a prior.
4. **Jorion Hyperparameter Portfolio:** Applies hyperparameters on the expected return vector's prior distribution (Jorion, 1986).
5. **Minimum Variance Portfolio:** Aims to minimize portfolio variance, focusing on the least risky portfolio irrespective of expected returns.

Table 1

Performance metrics for various strategies with a portfolio size of 10, between January 2006 and June 2023. The best and worst values in each row are highlighted in green and red, respectively.

	Value-Weighted	Equally-Weighted	Log-Normal $\kappa = 1$	Log-Normal $\kappa = 2$	Black–Litterman	Jorion Hyperpar.	Shrinkage	Min. Variance
Cum. return	470.277%	323.815%	510.298%	539.170%	437.268%	405.787%	478.609%	243.692%
CAGR	10.463%	8.604%	10.892%	11.185%	10.087%	9.707%	10.554%	7.311%
Sharpe ratio	0.504	0.452	0.521	0.534	0.503	0.478	0.529	0.436
Prob. Sharpe ratio	98.266%	97.039%	98.500%	98.695%	98.239%	97.668%	98.619%	96.602%
Sortino ratio	0.726	0.638	0.738	0.757	0.723	0.673	0.747	0.624
Calmar ratio	0.205	0.167	0.227	0.246	0.205	0.185	0.211	0.180
Max. DD	−51.090%	−51.628%	−48.068%	−45.436%	−49.228%	−52.476%	−50.040%	−40.660%
Avg. loss	−0.934%	−0.858%	−0.953%	−0.917%	−0.890%	−0.918%	−0.860%	−0.691%
Avg. return	0.049%	0.041%	0.051%	0.052%	0.047%	0.046%	0.048%	0.033%
Avg. win	0.874%	0.800%	0.891%	0.865%	0.831%	0.872%	0.815%	0.676%
Best day	11.897%	10.879%	13.498%	16.664%	11.308%	12.454%	13.708%	10.739%
Worst day	−10.946%	−10.135%	−13.319%	−16.152%	−10.379%	−12.948%	−12.415%	−8.575%
Ann. Vol.	22.258%	20.011%	22.361%	22.228%	21.170%	21.917%	20.801%	16.565%
Daily VaR	−2.257%	−2.033%	−2.266%	−2.251%	−2.147%	−2.225%	−2.107%	−1.683%
Avg. turnover	4.411%	7.573%	6.619%	9.794%	4.314%	21.248%	16.945%	14.707%

6. Shrinkage Portfolio: Utilizes a shrinkage approach towards the covariance matrix estimation to improve the efficiency of portfolio optimization (Ledoit and Wolf, 2004).

Full code implementations are available at <https://github.com/vilnik/positive-tangency-portfolio-weights>. The new method employs the PyMC package in Python and the No U-Turn Sampler (NUTS) to conduct MCMC simulations, utilizing ϵ to the power of the posterior mean of $\log(v)$ to ascertain the optimal portfolio weights (Hoffman et al., 2014; Abril-Pla et al., 2023). The Black–Litterman, Jorion Hyperparameter, and Shrinkage portfolios employ the PyPortfolioOpt Python package for performing portfolio optimization, adhering to (2.4) and imposing positive weight constraints (Martin, 2021). In contrast, the Minimum Variance Portfolio, while also utilizing PyPortfolioOpt, diverges by focusing its objective function on variance minimization, in a manner similar to the approach described in Jagannathan and Ma (2003). To maintain comparability among the methods, we carefully select risk aversion coefficients to ensure that the annualized volatility is approximately matched across different strategies (when applicable).

3.1. Backtesting results

We assess the strategies using a broad array of risk and return metrics, generated via the QuantStats Python package, accessible at <https://github.com/ranaroussi/quantstats>.

Given the high liquidity of the stocks, we apply a 3 bps turnover cost to reflect the minimal bid–ask spread. Our performance metrics, based on daily simple returns, provide a detailed risk profile. We use both traditional metrics like Sharpe and Sortino ratios, which consider excess returns, and others that use unadjusted returns. Furthermore, to enhance the reliability of Sharpe ratio comparisons, a key metric in our analysis, we employ the probabilistic Sharpe ratio with a zero benchmark, as recommended by Bailey and Lopez de Prado (2012). This method surpasses traditional hypothesis testing, offering a robust approach to the inherent uncertainty in Sharpe ratio estimation.

Tables 1 and 2 present the performance metrics for portfolio sizes of 10 and 20, respectively, providing insights into the efficacy of each strategy.

The Log-Normal Prior method, both with $\kappa = 1$ and $\kappa = 2$, demonstrates superior performance across multiple metrics for both portfolio sizes. Specifically, with a portfolio size of 10, this strategy achieves a cumulative return of 539.170% for $\kappa = 2$, outperforming all other methods. The Log-Normal Prior method also shows strong risk-adjusted performance with the highest Sharpe and Sortino ratios for both portfolio sizes. It is noteworthy, however, that this method exhibits susceptibility to sharp downturns, particularly marked in the worst day performance with $\kappa = 2$. Moreover, the advantage of the Log-Normal Prior method over other strategies seems to be more pronounced for a portfolio size of 10 than for 20.

In addition to the Log-Normal Prior method, the Value-Weighted and Black–Litterman portfolios demonstrate promising results in certain metrics. Conversely, the Equally-Weighted, Jorion Hyperparameter and Shrinkage approaches tend to underperform in comparison. It is also important to note the effective performance of the Minimum Variance portfolio in achieving its goal, demonstrated by its low annual volatility and daily VaR in both portfolio sizes.

Figs. 1 and 2 supplement with visual representations of returns, drawdown and rolling volatility.

Our visual analyses reveal consistent patterns across the various methods. Notably, the Black–Litterman portfolio closely mirrors the Value-Weighted Portfolio. This similarity is anticipated, as we do not impose absolute or relative views, thus granting significant influence to the prior value-weighted portfolio. The approach utilizing the Log-Normal Prior inherently integrates both the prior and observed data, resulting in a portfolio that reflects a balance of these elements. It is also important to observe that the portfolios informed by the Jorion Hyperparameter and Minimum Variance methods, and especially the Shrinkage method, display nuanced

Table 2

Performance metrics for various strategies with a portfolio size of 20, between January 2006 and June 2023. The best and worst values in each row are highlighted in green and red, respectively.

	Value-Weighted	Equally-Weighted	Log-Normal $\kappa = 1$	Log-Normal $\kappa = 2$	Black-Litterman	Jorion Hyperpar.	Shrinkage	Min. Variance
Cum. return	413.721%	315.687%	432.677%	439.548%	386.048%	325.988%	273.239%	279.066%
CAGR	9.805%	8.484%	10.033%	10.113%	9.458%	8.636%	7.818%	7.914%
Sharpe ratio	0.492	0.454	0.503	0.507	0.491	0.440	0.407	0.485
Prob. Sharpe ratio	98.008%	97.076%	98.200%	98.259%	97.986%	96.663%	95.497%	97.850%
Sortino ratio	0.703	0.640	0.712	0.715	0.701	0.616	0.568	0.688
Calmar ratio	0.185	0.158	0.198	0.205	0.185	0.138	0.124	0.196
Max. DD	-52.973%	-53.682%	-50.621%	-49.344%	-51.074%	-62.491%	-62.853%	-40.322%
Avg. loss	-0.892%	-0.818%	-0.901%	-0.887%	-0.849%	-0.899%	-0.870%	-0.664%
Avg. return	0.046%	0.040%	0.047%	0.047%	0.044%	0.042%	0.039%	0.035%
Avg. win	0.823%	0.773%	0.825%	0.815%	0.783%	0.859%	0.815%	0.627%
Best day	11.030%	11.018%	12.025%	13.705%	10.357%	13.398%	13.283%	10.276%
Worst day	-11.222%	-10.922%	-12.661%	-14.168%	-10.638%	-13.602%	-13.068%	-9.211%
Ann. Vol.	21.109%	19.467%	21.003%	20.973%	20.071%	21.212%	20.927%	15.800%
Daily VaR	-2.141%	-1.977%	-2.129%	-2.126%	-2.036%	-2.156%	-2.130%	-1.602%
Avg. turnover	3.236%	6.430%	4.669%	6.912%	3.188%	25.564%	20.272%	17.799%

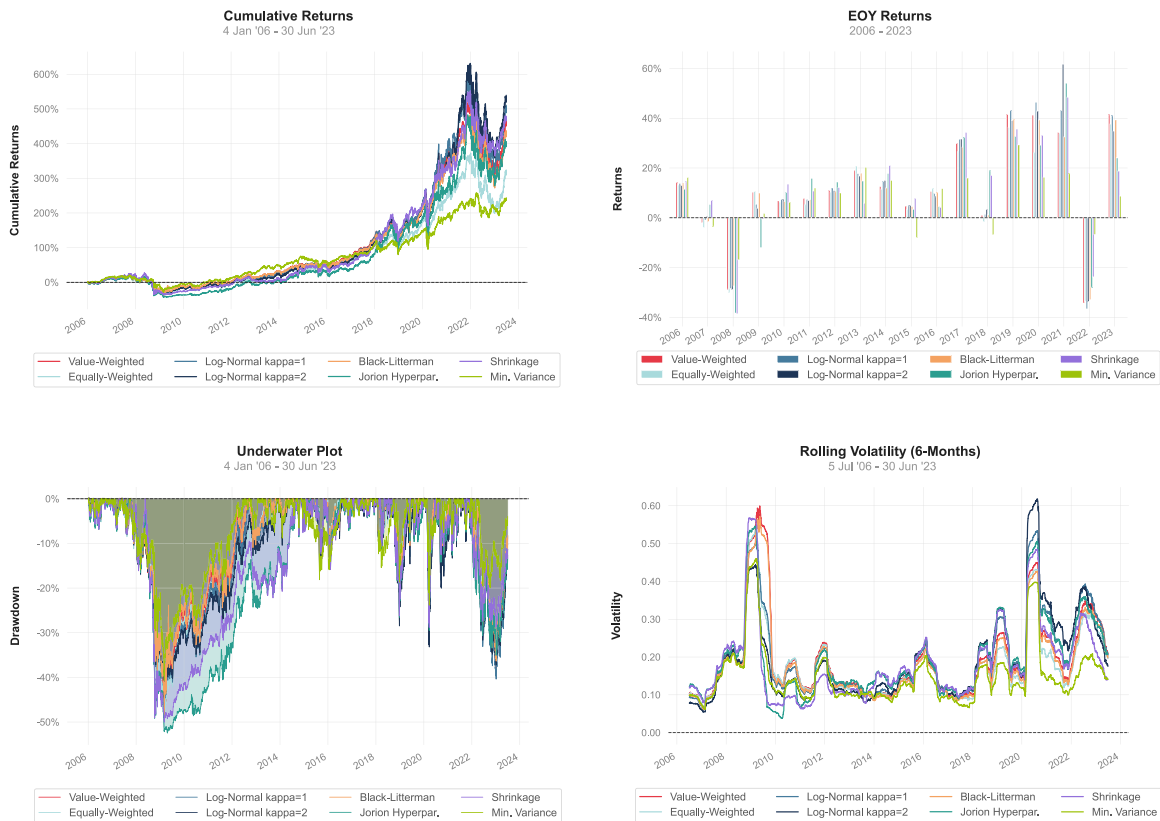


Fig. 1. Visual representation of returns, drawdown and rolling volatility for various strategies with a portfolio size of 10, between January 2006 and June 2023.

departures from the other strategies. Such variations could be attributed to these methods' lack of a reference portfolio, a feature central to both the Log-Normal Prior and Black-Litterman methods.

Although visual outcomes appear largely congruent, the underlying mechanics of our pioneering approach set it apart. It distinguishes itself by facilitating direct and constrained inference on portfolio weights, ensuring strictly positive values. In stark contrast, the conventional techniques derive optimal weights indirectly by integrating estimates of μ and Σ within a convex optimization framework. Such an approach often obfuscates clear insights into the nuances of optimal weight determination. By modeling ν directly, our method crystallizes understanding and provides a granular perspective on the inherent uncertainties of

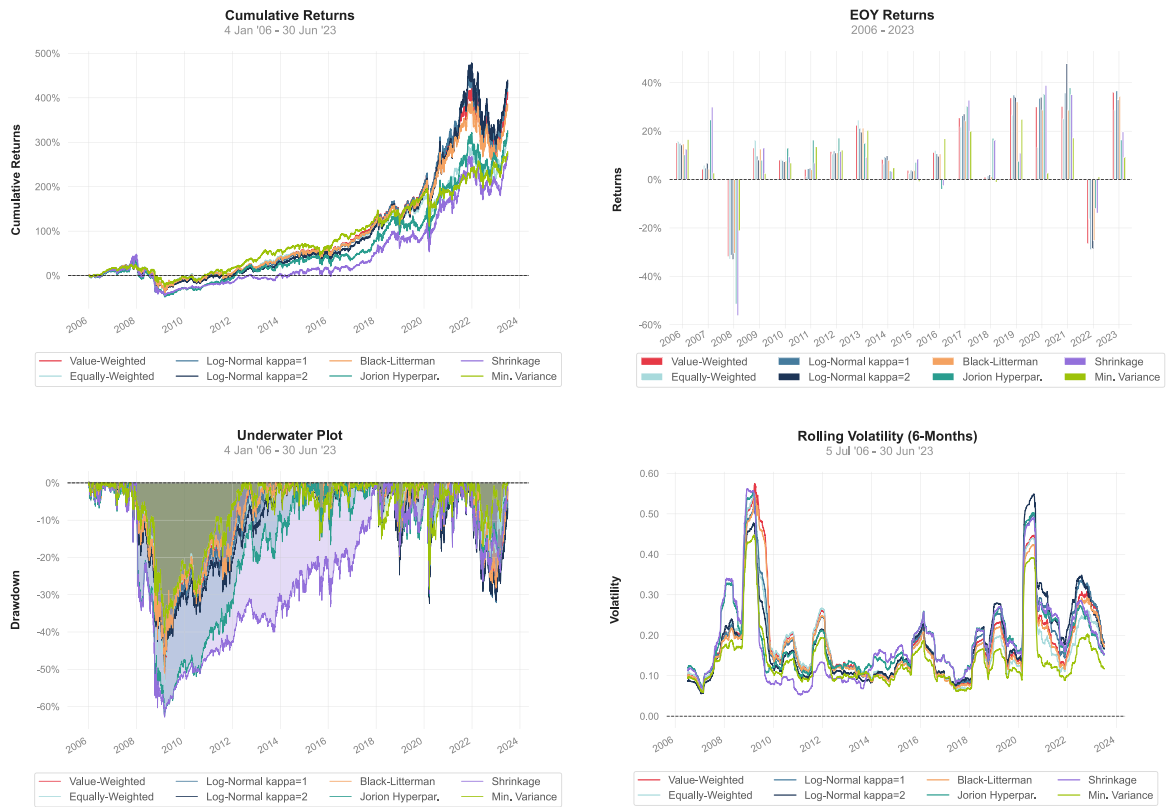


Fig. 2. Visual representation of returns, drawdown and rolling volatility for various strategies with a portfolio size of 20, between January 2006 and June 2023.



Fig. 3. Posterior distributions of log ν with a portfolio size of 10 during the June 2023 rebalancing, with the left figure representing $\kappa = 1$ and the right figure representing $\kappa = 2$.

these weights. Figs. 3 and 4 illustrate the posterior distribution of log portfolio weights, compared with the preset value-weighted prior during the June 2023 rebalancing. The use of log weights, as opposed to regular weights, provides a clearer visualization of the posterior distribution.

Upon examining Figs. 3 and 4, a clear distinction emerges in the spread of the posterior distributions for different values of κ . Specifically, when $\kappa = 2$, the spread of the posterior distributions appears slightly broader than when $\kappa = 1$. This observation aligns with our theoretical expectations. A larger κ indicates a prior with a higher variance, and thus it is consistent to observe a more expansive spread in the posterior distributions in such cases.

A deviation from the value-weighted prior or a wider spread can indicate heightened uncertainty about a particular stock. Such deviations necessitate a more comprehensive analysis to understand the underlying factors. Moreover, it is worth noting that in our approach, we do not strictly mandate the portfolio weights to sum up to one. Any difference from unity suggests an amount that

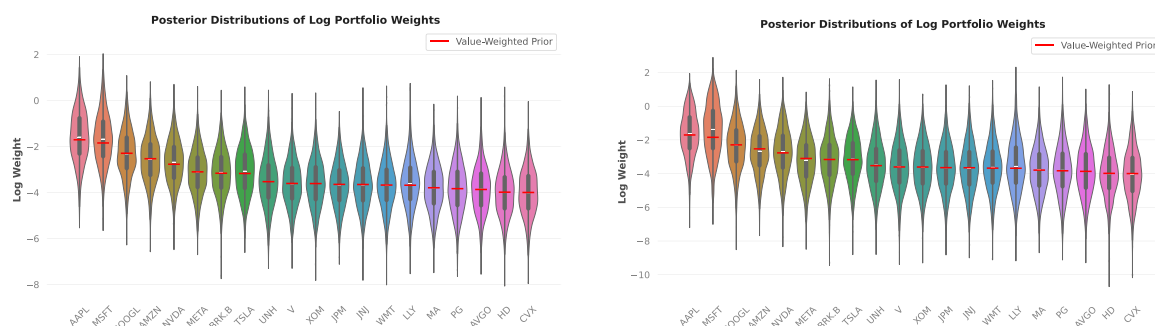


Fig. 4. Posterior distributions of log ν with a portfolio size of 20 during the June 2023 rebalancing, with the left figure representing $\kappa = 1$ and the right figure representing $\kappa = 2$.

is either borrowed or invested in the risk-free asset. This flexibility is a significant component of our model, offering an additional layer of nuance in portfolio optimization.

4. Conclusion

We introduced an innovative Bayesian methodology for constructing tangency portfolios, specifically tailored to environments with short-selling constraints. Central to our approach is a novel parametrization of asset returns, complemented by a distinctive prior. Employing MCMC techniques and real market data, empirical results affirm the superior performance of our methodology relative to existing methods. Notably, our approach adeptly melds prior beliefs with empirical observations, fostering robust and insightful inferences about portfolio weights. Such granularity empowers investors to make informed investment choices and pinpoint assets where discrepancies between prior beliefs and observed data are most pronounced. Future research avenues could explore the efficacy of our method across larger portfolios, elongated time frames, and with additional portfolio constraints.

CRedit authorship contribution statement

Olha Bodnar: Writing – review & editing, Writing – original draft, Methodology, Conceptualization. **Taras Bodnar:** Writing – review & editing, Writing – original draft, Supervision, Methodology, Conceptualization. **Vilhelm Niklasson:** Writing – review & editing, Writing – original draft, Visualization, Software, Methodology, Formal analysis, Data curation, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data collection details and code are documented in the GitHub repository linked in the manuscript, ensuring transparency and reproducibility.

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